

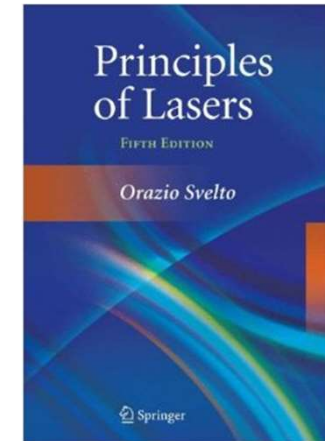


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch2-2 Open Cavity & Gaussian Beam



Dr. Dan Luo(罗丹)

Email: luod@sustech.edu.cn

Office: Room 331 at College
of Engineering South Tower

Tel: 0755-88018552



Chapter 2: Open Cavity & Gaussian Beam

❖ 2.3 Mode for Open Cavity & Diffraction Theory

- Introduction
- The self-reconstruction of mode
- Fresnel Kirchhoff's law
- The physical meaning of the integral equation

❖ 2.5 The self-reconstructed mode of square confocal cavity

- Variables separation
- Solutions: Analytical solution and approximation solution

1. Introduction

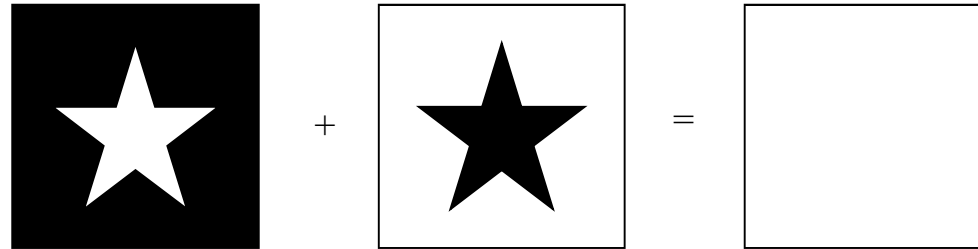
Diffraction plays an important role in spatial distribution of laser energy

After multiple traveling, a steady-state field will be formed: energy distribution will not be influenced by diffraction any more; the field of light is the same as the field before previous reflection

- The shape of the field is the same
- The amplitude is different
- The phase is difference

Babinet's Principle: In physics, Babinet's principle is a theorem concerning diffraction. It mentions that the diffraction pattern from an opaque body is identical to that from a hole of the same size and shape except for the overall forward beam intensity.

Complementary (互补)

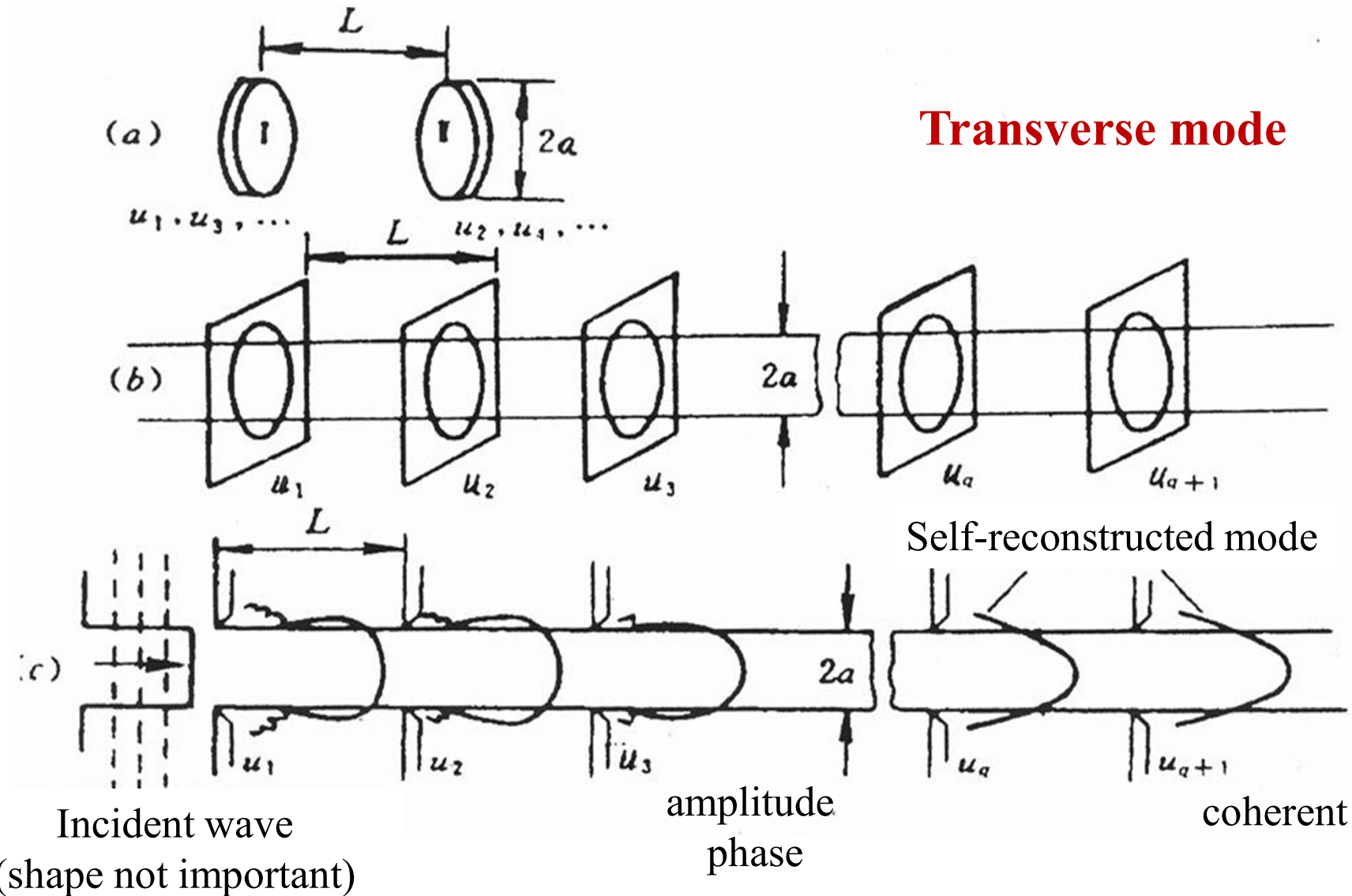


$$-\frac{i}{\lambda} \iint_{\Sigma_A} U_0(Q) \frac{e^{ikr}}{r} d\Sigma - \frac{i}{\lambda} \iint_{\Sigma_B} U_0(Q) \frac{e^{ikr}}{r} d\Sigma = -\frac{i}{\lambda} \iint_{\Sigma_0} U_0(Q) \frac{e^{ikr}}{r} d\Sigma$$

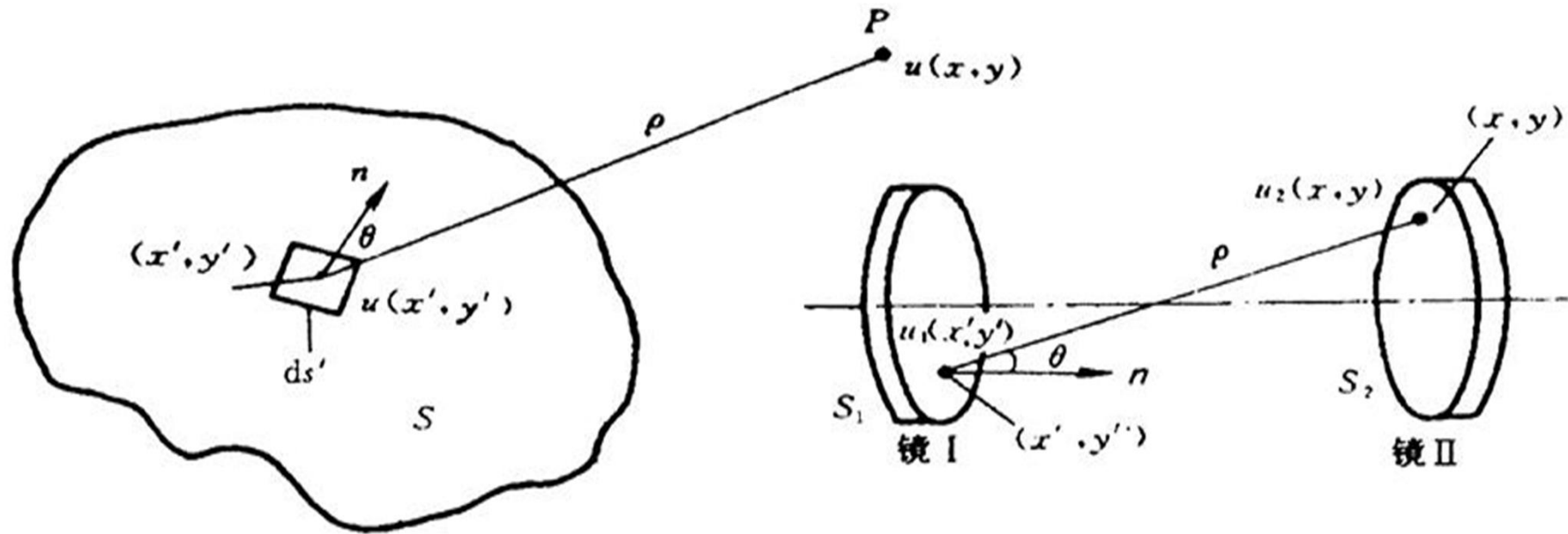
$$U_A(P) + U_B(P) = U_0(P)$$

The far-field diffraction patterns induced by a pair of complementary structures have the same shape

2. The Self-Reconstruction of Mode



3. Fresnel Kirchhoff's Law



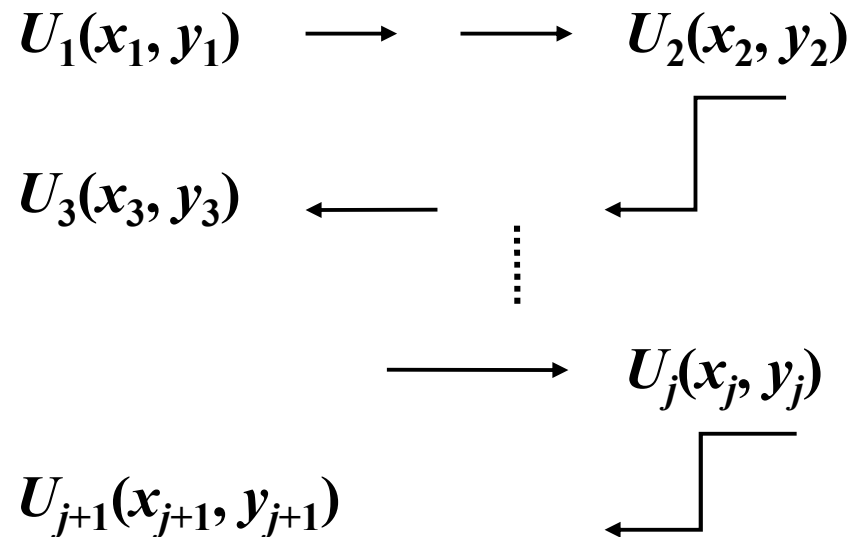
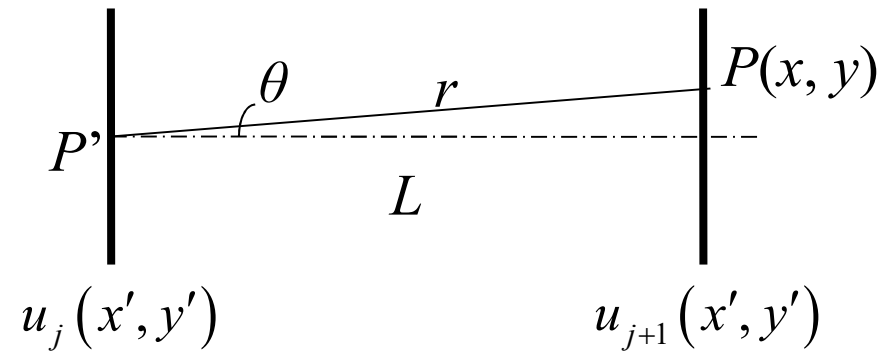
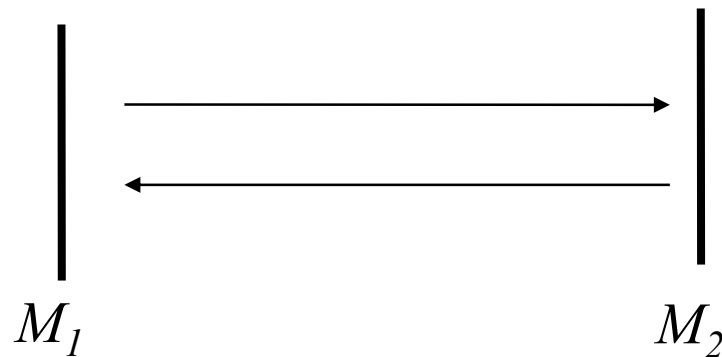
$$u(x, y) = \frac{ik}{4\pi} \iint_S u(x', y') \frac{e^{-ik\rho}}{\rho} (1 + \cos \theta) ds'$$

Light field distribution
function on the surface S

Spherical wave
from each point
source

Angle-related
factor

The idea of self-reconstruction of mode



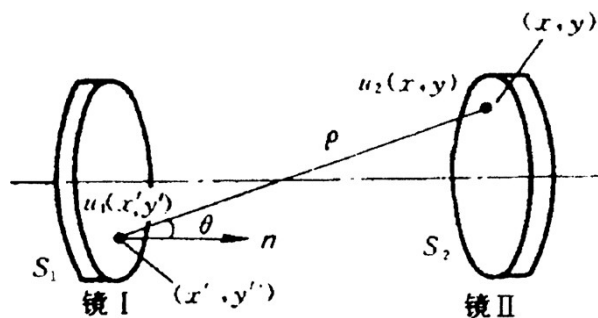
The same light field

The self-reconstruction of mode

With multiple travels, beside the decrease of amplitude and change of phase, the light field U_{j+1} can construct U_j

Condition for reconstructed mode

$$U_{j+1}(x, y) = \frac{1}{\gamma} U_j(x, y)$$



- Fresnel Kirchhoff's law
- self-reconstruction of mode

The integral equation of self-reconstruction field in optical cavity

$$u_{j+1}(x, y) = \frac{ik}{4\pi} \iint_S u_j(x', y') \frac{e^{-ik\rho}}{\rho} (1 + \cos \theta) ds'$$

The integral equation

$$u_j(x, y) = \gamma \frac{ik}{4\pi} \iint_S u_j(x', y') \frac{e^{-ik\rho}}{\rho} (1 + \cos \theta) ds'$$

$$u_{j+1}(x, y) = \gamma \frac{ik}{4\pi} \iint_S u_{j+1}(x', y') \frac{e^{-ik\rho}}{\rho} (1 + \cos \theta) ds'$$

.....

.....

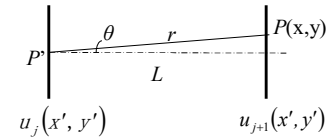
The integral of the self-reconstructed mode

$$V(x, y) = \gamma \frac{ik}{4\pi} \iint V(x', y') \frac{e^{-ik\rho}}{\rho} (1 + \cos \theta) ds'$$

The stable field distribution in cavity

The integral of the self-reconstructed mode

$$V(x, y) = \gamma \frac{ik}{4\pi} \iint V(x', y') \frac{(1 + \cos \theta)}{\rho} e^{-ik\rho} ds'$$



$$\because L \gg x, y \quad \rho \approx L, \cos \theta \approx 1 \Rightarrow (1 + \cos \theta) / \rho \approx 2 / L$$

$$V(x, y) = \gamma \frac{i}{\lambda L} \iint V(x', y') e^{-ik\rho(x, y, x', y')} ds'$$



$$V(x, y) = \gamma \iint K(x, y, x', y') V(x', y') ds'$$

where $K(x, y, x', y') = \frac{i}{\lambda L} e^{-ik\rho(x, y, x', y')}$ Kernel (核) of the integral equation

- Capable for any symmetric cavity (Parallel plane, confocal, general spherical mirror cavity)
- Find $V(x, y)$ $\Leftrightarrow$ mode distribution $\Rightarrow$ Solve the Integral Equation

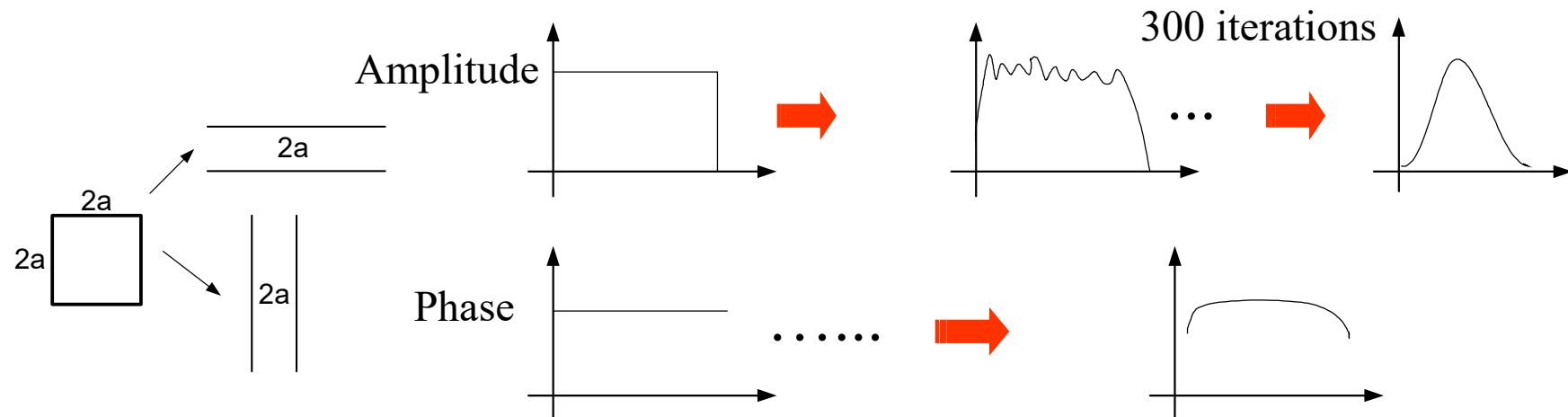
Solve the integral of the self-reconstructed mode?

1. The analytical solution: only for symmetrical confocal cavity

Type of Confocal Cavity	Analytical Solution	Approximation
Square	Prolate-ellipsoid function	Hermite-Gaussian function
Circular	Hyper-ellipsoid function	Laguerre-Gaussian function

The solution for other stable spherical mirror cavities can be obtained by means of "equivalent" to the symmetry confocal cavity.

2. The numerical solution: numerical iteration method (迭代)



4. Integral equation: physical meaning

$$V(x, y) = \gamma \frac{i}{\lambda L} \iint V(x', y') e^{-ik\rho(x, y, x', y')} ds'$$

Eigen Function 本征函数	Eigen Value 本征值
-------------------------------	---------------------------

1) The expression of Eigen function

$$V_{mn}(x, y) = A_{mn}(x, y) e^{i\varphi_{mn}(x, y)}$$

$V_{mn}(x, y)$ Field distribution on the mirror (Eigen function $\Leftrightarrow$ transverse mode)

$A_{mn}(x, y)$ Amplitude distribution on the mirror

$\varphi_{mn}(x, y)$ Phase distribution on the mirror

- There is no analytical solution for general spherical mirror cavity
- Analytical solution can be obtained only for symmetrical confocal cavity

2) The meaning of eigen value γ — complex constant

$$\gamma_{mn} = |\gamma_{mn}| e^{i\delta\varphi_{mn}} \quad \text{— Transmission factor} \quad \gamma \text{ is related to } \delta_D$$

$$\text{if } \gamma_{mn} = e^{\alpha+i\beta} \quad u_{j+1} = \frac{1}{\gamma_{mn}} u_j \Rightarrow u_{j+1} = \underbrace{(e^{-\alpha} u_j)}_{\substack{\uparrow \\ \text{decrease of amplitude per round}}} e^{-i\beta} \quad \nwarrow \text{phase shift}$$

$$\delta_D \equiv \frac{|u_{mnj}|^2 - |u_{mnj+1}|^2}{|u_{mnj}|^2} = 1 - e^{-2\alpha} = 1 - \frac{1}{|\gamma_{mn}|^2}$$

δ_D — The relative power loss per travel — 单程损耗

$|\gamma_{mn}|$ — The single-pass loss of self-reconstructed mode

— Different transverse mode has different δ_D

— $|\gamma_{mn}| \uparrow \rightarrow$ 模的单程损耗 $\delta_D \uparrow$

3) The phase shift $\delta\varphi_{mn}$ — the total phase shift of the self reconstructed modes

$$\delta\varphi_{mn} = \arg u_{j+1} - \arg u_j \quad \because \gamma = e^{\alpha+i\beta} \quad u_{j+1} = \frac{1}{\gamma} u_j$$

$$\delta\varphi_{mn} = -\beta = \arg \frac{1}{\gamma}$$

Geometrical phase shift
几何相移

$$\delta\varphi_{mn} = -\beta = -kL + \Delta\varphi = -\frac{2\pi L}{\lambda} + \underline{\Delta\varphi}$$

Additional phase shift
附加相移

Resonant condition $\delta\varphi_{mn} = q\pi$ If we know γ_{mn} , can obtain the resonant frequency of the reconstructed mode

The modulus of the complex constant is the one-round loss of the self reconstructed mode, and its argument measures the phase shift, which also determines the resonant frequency of the mode.

Variable separation

$$V(x, y) = \gamma \iint K(x, y, x', y') V(x', y') ds'$$

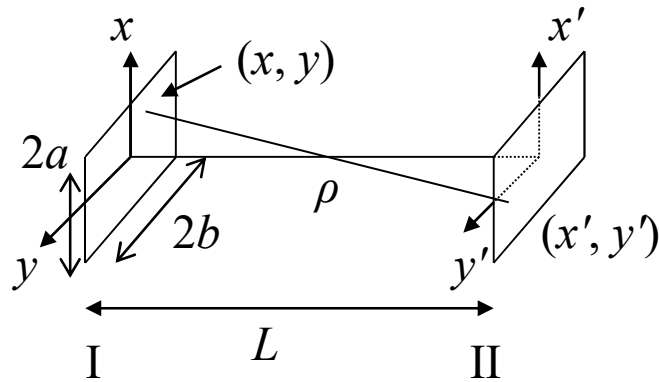
$$\text{When } L, R \gg a \quad K(x, y, x', y') = \frac{i}{\lambda L} e^{-ik\rho(x, y, x', y')}$$

$$\text{if } \rho(x, y, x', y') \stackrel{?}{=} \rho_x(x, x') \rho_y(y, y')$$

$$K(x, y, x', y') = \frac{i}{\lambda L} e^{-ik\rho(x, y, x', y')} \Rightarrow K(x, y, x', y') = K_x(x, x') K_y(y, y')$$

$$\begin{array}{l} \text{assume} \\ V(x, y) = V(x) V(y) \end{array} \quad \left\{ \begin{array}{l} V(x) = \gamma_x \int_{-a}^{+a} K_x(x, x') V(x') dx' \\ V(y) = \gamma_y \int_{-b}^{+b} K_y(y, y') V(y') dy' \end{array} \right.$$

1) Rectangular Planar Cavity



$$\rho = \sqrt{(x - x')^2 + (y - y')^2 + L^2}$$

$$\rho = \sqrt{(x - x')^2 + (y - y')^2 + L^2} = L \sqrt{\left(\frac{x - x'}{L}\right)^2 + \left(\frac{y - y'}{L}\right)^2 + 1}$$

$$\approx L \left[1 + \frac{1}{2} \left(\frac{x - x'}{L}\right)^2 + \frac{1}{2} \left(\frac{y - y'}{L}\right)^2 - \frac{1}{8} \left(\frac{x - x'}{L}\right)^4 - \frac{1}{8} \left(\frac{y - y'}{L}\right)^4 - \frac{1}{4} \left(\frac{x - x'}{L}\right)^2 \left(\frac{y - y'}{L}\right)^2 + \dots \right]$$

$$(1+x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha+1)\dots(\alpha-n+1)}{n!} x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots + \frac{\alpha(\alpha+1)\dots(\alpha-n+1)}{n!} x^n + \dots, x \in (-1, 1)$$

$$\rho = \sqrt{(x-x')^2 + (y-y')^2 + L^2} = L \sqrt{\left(\frac{x-x'}{L}\right)^2 + \left(\frac{y-y'}{L}\right)^2 + 1}$$

$$\approx L \left[1 + \frac{1}{2} \left(\frac{x-x'}{L}\right)^2 + \frac{1}{2} \left(\frac{y-y'}{L}\right)^2 - \frac{1}{8} \left(\frac{x-x'}{L}\right)^4 - \frac{1}{8} \left(\frac{y-y'}{L}\right)^4 - \frac{1}{4} \left(\frac{x-x'}{L}\right)^2 \left(\frac{y-y'}{L}\right)^2 + \dots \right]$$

$$\Rightarrow e^{-ik\rho} \approx e^{-ikL \left[1 + \frac{1}{2} \left(\frac{x-x'}{L}\right)^2 + \frac{1}{2} \left(\frac{y-y'}{L}\right)^2 \right]} = e^{-ikL} e^{-ik \left[\frac{(x-x')^2}{2L} + \frac{(y-y')^2}{2L} \right]}$$

$$\Rightarrow K(x, y, x', y') = \frac{i}{\lambda L} e^{-ik\rho(x, y, x', y')} = \sqrt{\frac{i}{\lambda L}} e^{-ikL} e^{-ik \frac{(x-x')^2}{2L}} \sqrt{\frac{i}{\lambda L}} e^{-ikL} e^{-ik \frac{(y-y')^2}{2L}}$$

$$V(x, y) = \gamma \iint K(x, y, x', y') V(x', y') ds'$$

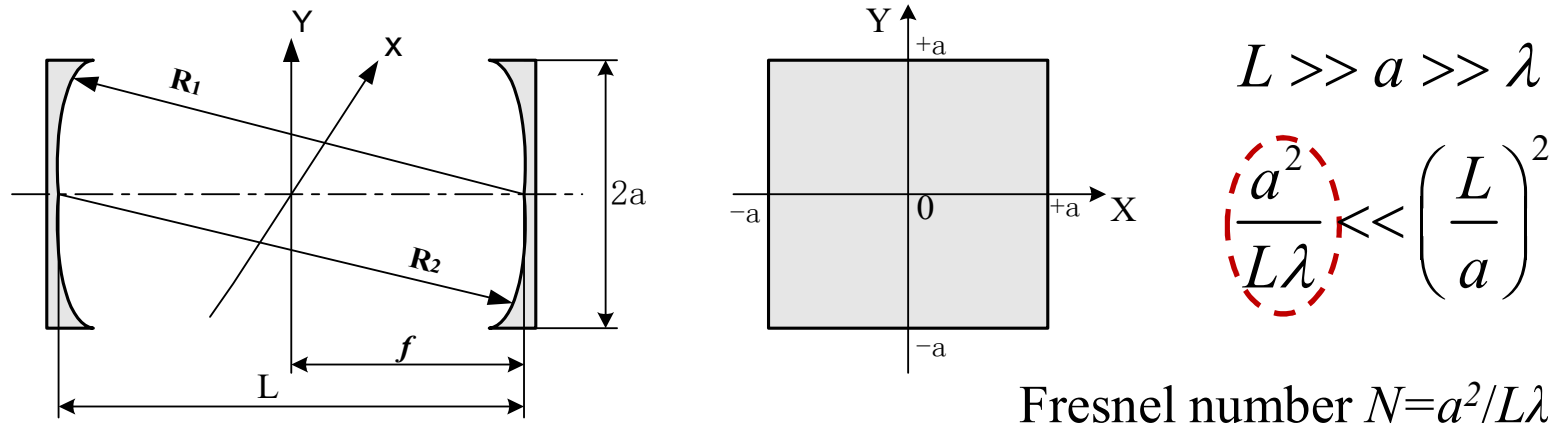
$$\left. \begin{aligned} K_x(x, x') &= \sqrt{\frac{i}{\lambda L}} e^{-ikL} e^{-ik\frac{(x-x')^2}{2L}} \\ K_y(y, y') &= \sqrt{\frac{i}{\lambda L}} e^{-ikL} e^{-ik\frac{(y-y')^2}{2L}} \end{aligned} \right\} \Rightarrow \left\{ \begin{aligned} V(x) &= \gamma_x \int_{-a}^{+a} K_x(x, x') V(x') dx' \\ V(y) &= \gamma_y \int_{-b}^{+b} K_y(y, y') V(y') dy' \\ \gamma_x \gamma_y &= \gamma \end{aligned} \right.$$

More general expression

V_m and V_n are the m^{th} and n^{th} solutions, γ_m and γ_n the corresponding complex constants:

$$\left. \begin{aligned} V_m(x) &= \gamma_m \int_{-a}^{+a} K_x(x, x') V_m(x') dx' \\ V_n(y) &= \gamma_n \int_{-b}^{+b} K_y(y, y') V_n(y') dy' \end{aligned} \right\} \Rightarrow \left\{ \begin{aligned} V_{mn}(x, y) &= V_m(x) \cdot V_n(y) \\ \gamma_{mn} &= \gamma_m \gamma_n \end{aligned} \right.$$

2) Square Confocal Cavity



Variables separation (分离变量)

$$V(x, y) = \gamma \frac{i}{\lambda L} \iint V(x', y') e^{-ik\rho(x, y, x', y')} ds'$$

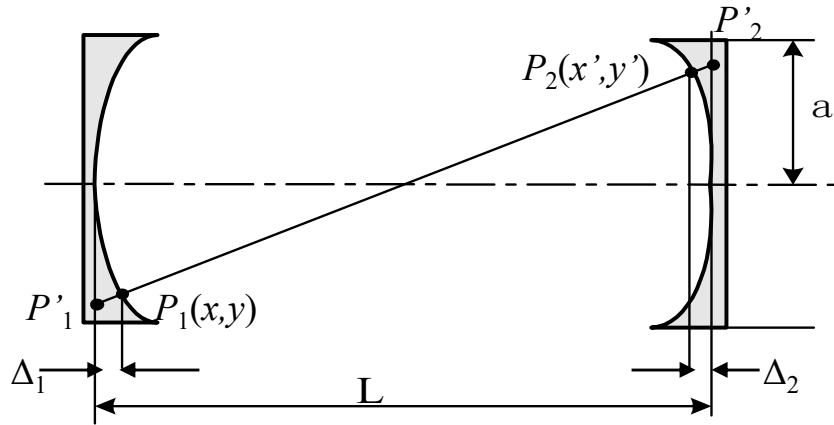
$$V(x, y) = V(x)V(y) \begin{cases} V(x) = \gamma_x \int_{-a}^{+a} K_x(x, x') V(x') dx' \\ V(y) = \gamma_y \int_{-a}^{+a} K_x(y, y') V(y') dy' \end{cases}$$

$$V(x, y) = \gamma \iint K(x, y, x', y') V(x', y') ds'$$

$$K(x, y, x', y') = \frac{i}{\lambda L} e^{ik\rho(x, y, x', y')}$$

$$\rho(x, y, x', y') = \rho_x(x, x') \rho_y(y, y')$$

$$\Rightarrow K(x, y, x', y') = K(x, x') K(y, y')$$



$$\rho = \bar{P}'_1 \bar{P}'_2 - \bar{P}'_1 \bar{P}_1 - \bar{P}'_2 \bar{P}_2$$

$$\bar{P}'_1 \bar{P}'_2 \approx L + \frac{(x - x')^2}{2L} + \frac{(y - y')^2}{2L}$$

$$\bar{P}'_1 \bar{P}_1 \approx \Delta_1 \approx \frac{x^2 + y^2}{2R_1} \quad \bar{P}'_2 \bar{P}_2 \approx \Delta_2 \approx \frac{x'^2 + y'^2}{2R_2}$$

$$\begin{aligned} \rho(x, y, x', y') &= L + \frac{1}{2L} \left[\left(1 - \frac{L}{R_1} \right) (x^2 + y^2) + \left(1 - \frac{L}{R_2} \right) (x'^2 + y'^2) - 2(xx' + yy') \right] \\ &= L + \frac{1}{2L} \left[g_1 (x^2 + y^2) + g_2 (x'^2 + y'^2) - 2(xx' + yy') \right] \end{aligned}$$

for symmetrical
confocal cavity

$$R_1 = R_2 = R = L$$

$$\rho(x, y, x', y') = L - \frac{1}{L} (xx' + yy')$$

$$V(x, y) = \gamma \frac{i}{\lambda L} \iint_s V(x', y') e^{-ik\rho(x, y, x', y')} ds'$$

Square confocal cavity  $\rho = L - (xx' + yy')/L$

$$V_{mn}(x, y) = \gamma_{mn} \left(\frac{i}{\lambda L} e^{-ikL} \right) \int_{-a}^a \int_{-a}^a V_{mn}(x', y') e^{ik \frac{xx' + yy'}{L}} dx' dy'$$

Separate $V_{mn} = F_m(X)G_n(Y)$  $X = \frac{\sqrt{c}}{a}x, \quad Y = \frac{\sqrt{c}}{a}y$ $c = 2\pi N = 2\pi \left(\frac{a^2}{L\lambda} \right)$  Fresnel number $c = 2\pi N$  $\gamma_{mn} = \frac{1}{\sigma_m \sigma_n}$

$$\sigma_m \sigma_n F_m(X) G_n(Y) = \frac{ie^{-ikL}}{2\pi} \int_{-\sqrt{c}}^{+\sqrt{c}} F_m(X') e^{iXX'} dX' \int_{-\sqrt{c}}^{+\sqrt{c}} G_n(Y') e^{iYY'} dY'$$

Separate into two independent equations $\left\{ \begin{array}{l} \sigma_m F_m(X) = \sqrt{\frac{ie^{-ikL}}{2\pi}} \int_{-\sqrt{c}}^{+\sqrt{c}} F_m(X') e^{iXX'} dX' \\ \sigma_n F_n(Y) = \sqrt{\frac{ie^{-ikL}}{2\pi}} \int_{-\sqrt{c}}^{+\sqrt{c}} F_n(Y') e^{iYY'} dY' \end{array} \right.$